High Returns, Low Investment: Public Capital and the Euler Equation Puzzle

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Abstract

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1 Introduction

Much of development economics research attempt to understand what is holding back economic development in lower- and middle-income countries. The categorisation of countries into 'developing' versus 'developed' implies that the former have higher development potentials. The Solow (1956) growth model predicts exactly this pattern because poorer economies with smaller capital stocks should face higher marginal returns to capital investment and therefore grow faster than richer economies. While the broad pattern of higher GDP growth rates in lower-income countries over the past four decades is consistent with this prediction, many countries have still believed to been growing at rates below their potentials.

Lucas (1990) and Caselli and Feyrer (2007) points out that the higher marginal product of capital in lower-income countries is often balanced-off by their lower endowment in complimentary factors like human capital, or lower total factor productivity. This provides an aggregate-level explanation for why capitals do not flow from higher-income to lower-income countries as much as expected. However, the puzzle is not fully addressed because on the aggregate-level, it does not explain why lower-income countries do not invest in human capital or R&D to benefit from the high capital returns. Moreover, through micro-level RCTs, de Mel et al. (2008) show that microenterprises in Sri Lanka face 4.6%-5.3% real returns to capital, and Balboni et al. (2022) show significant long-run returns on agricultural capitals sufficient to push households in Bangladesh out of a poverty trap. Kremer et al. (2019) uses an household-level intertemporal constrained optimisation model to illustrate an '*Euler Equation Puzzle*' where people in lower-income settings invest sub-optimally given high returns from capital investment.

Relating to this paradox, a large body of literature including Balboni et al. (2022) and research covered in Banerjee (2003) point to credit constraints facing households and firms; some including Acemoglu et al. (2001), Mauro (1995) and Banerjee et al. (2002) point to institutional ineffectiveness and insufficient property rights protections; while Kremer et al. (2019) point to studies in behavioural channels such as present bias. However, the existing literature focus primarily on private investment channels. For example, these are scenarios where farmers investing in a cattle or fertilisers and benefit fully from the output gains. Meanwhile, returns on certain capitals like infrastructure or human capital are also high in lower-income countries. Investment in public capitals often require fiscal spending from the government funded primarily by tax revenues. Besley and Persson (2013) show that in lower-income countries taxation is more difficult administratively while frictions against efficient spending are stronger.

Nonetheless, it might still seem irrational for above-subsistence households in lower-income countries to avoid taxes even with low expected penalties because the long-term returns from public capitals can be sufficiently high. This paper shows that the set-up of the taxation-to-government-spending mechanism can produce a collective action problem at the household level when perceived state taxation capacity is low.

In Section 2 of this paper, I will illustrate the collective action problem on public capital investment base on the intertemporal constrained optimisation model of Kremer et al. (2019).

2 A Household Intertemporal Constrained Optimisation Model Incorporating Public Capitals

I build on the intertemporal optimisation framework of Kremer et al. (2019). I first recall the model setup and present the Euler Equation Puzzle through the model. I then consider the extreme scenario where only public capitals exist and investment occurs only through taxation, showing that this generates a collective action problem and a government efficiency problem that dilutes the effective return to capital when state taxation capacity and efficiency is low. Finally, I will extend this analysis into a generalised model incorporating both public and private capital, allowing households to allocate investments between the two.

2.1 Baseline: Standard Euler Equation

Consider in a deterministic discrete-time intertemporal consumption model, households maximise their discounted lifetime utility over an infinite time horizon.

Optimisation Problem

Households face the following optimisation problem:

$$\max_{\{C_t\}} \sum_{t=0}^{\infty} \delta^t u(C_t), \tag{1}$$

where $0 < \delta < 1$ is the annual discount factor, C_t is the household consumption in period t, and $u(C_t)$ measures the utility from consuming C_t in period t. $u(C)$ is increasing and concave and follows Inada Conditions: $u'(C) > 0$, $u''(C) < 0$, $\lim_{C\to 0} u'(C) = \infty$, and $\lim_{C\to\infty} u'(C) = 0$.

Budget Constraint

$$x_{t+1} = f'(K_t)(x_t - C_t) \tag{2}$$

x_t is household assets ('cash on hand') in period t with x_0 given, C_t is consumption in period t, and $f'(K_t)$ is the gross return to capital. $f(K)$ is increasing and concave, following: $f'(K) > 0$, $f''(K) < 0$, $\lim_{K\to 0} f'(K) = \infty$, and $\lim_{K\to\infty} f'(K) = 1$ (long-run steady state).

Discrete Time Euler Equation

Solve for optimal intertemporal consumption path:

$$u'(C_t) = \delta f'(K_t) u'(C_{t+1}), \tag{3}$$

and assume constant-elasticity-of-substitution (CES) utility:

$$u(C) = \begin{cases} \frac{C^{1-\sigma}}{1-\sigma} & \text{if } \sigma \neq 1 \\ \ln(C) & \text{if } \sigma = 1 \end{cases}$$

so that the elasticity of intertemporal substitution is constant and independent of C_t. This gives the consumption growth rate:

$$\frac{C_{t+1}}{C_t} - 1 = (\delta f'(K_t))^{\frac{1}{\sigma}} - 1, \tag{4}$$

where $\frac{1}{\sigma}$ is the elasticity of intertemporal substitution.

The Puzzle

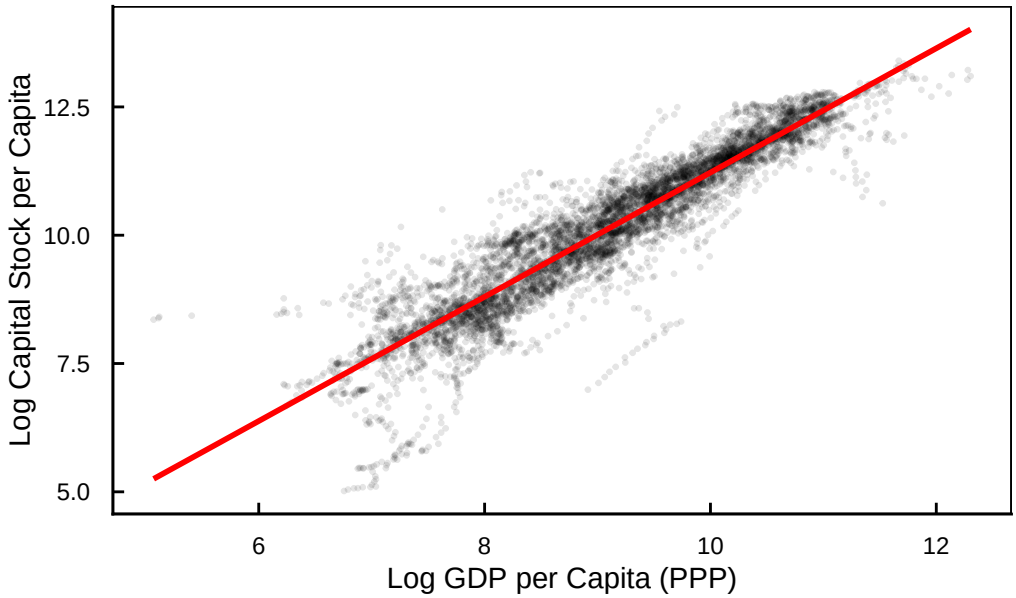
Assume that the model governs the intertemporal consumption behaviour of a representative household, the model should also apply at the macro level. To illustrate the puzzle with macro-level data, I use $\delta = 0.96$ [1]and $\sigma = 2$ [2] proposed in Kremer et al. (2019), and internal rates of return from the Penn World Table (PWT) [3] (Feenstra et al., 2015) as a proxy for $f'(K)$[4] to model the predicted consumption growth rates under Equation 4. Intuitively, a concave production function means that countries with lower capital stock face higher returns on capital investment. Figure 1 shows that countries with lower income have lower capital stock. Therefore, according to Equation 4, consumption growth rates should be high. However, Figure 2 Panel (A) shows that in many countries, the actual growth rates fall below the model predictions. Moreover, Figure 2 Panel (B) shows a confusing trend where countries with higher predicted consumption growth rates tend to have larger gap between predicted and actual rates. This pattern is most pronounced in countries such as Egypt, Venezuela, Côte d'Ivoire, and Zimbabwe. Therefore, the puzzle is why households in these economies invest at levels far below what the standard model predicts as optimal, and why is this pattern more pronounced in countries where returns on capitals are high.

2.2 Capital as a Public Good

The baseline model in Section 2.1 assumes capital investments are private for households. This means that households receive the full returns $f'(K)$ because they would own the invested capital in productions. However, capitals with public good features like infrastructure often instead require public investment commonly through government spending, and the benefits from these investments are shared among the population. Aschauer (1989) estimates a high average elasticity of 0.39 between output and public investment in the United States over 1949-1985, while Bom and Ligthart (2014) proposes a more modest estimate of 0.106. The latter estimate that the short-run national-level elasticity of 0.083 still imply a 16% marginal return to public capital in OECD

[1] $\frac{1}{\delta} - 1 \approx 0.04$, implying a 4% annual rate of time preference.
[2] Elasticity of intertemporal substitution of 0.5.
[3] Available for download at www.ggdc.net/pwt
[4] See https://www.rug.nl/ggdc/docs/the_next_generation_of_the_penn_world_table.pdf

Figure 1: Lower Capital Stock in Lower-Income Countries



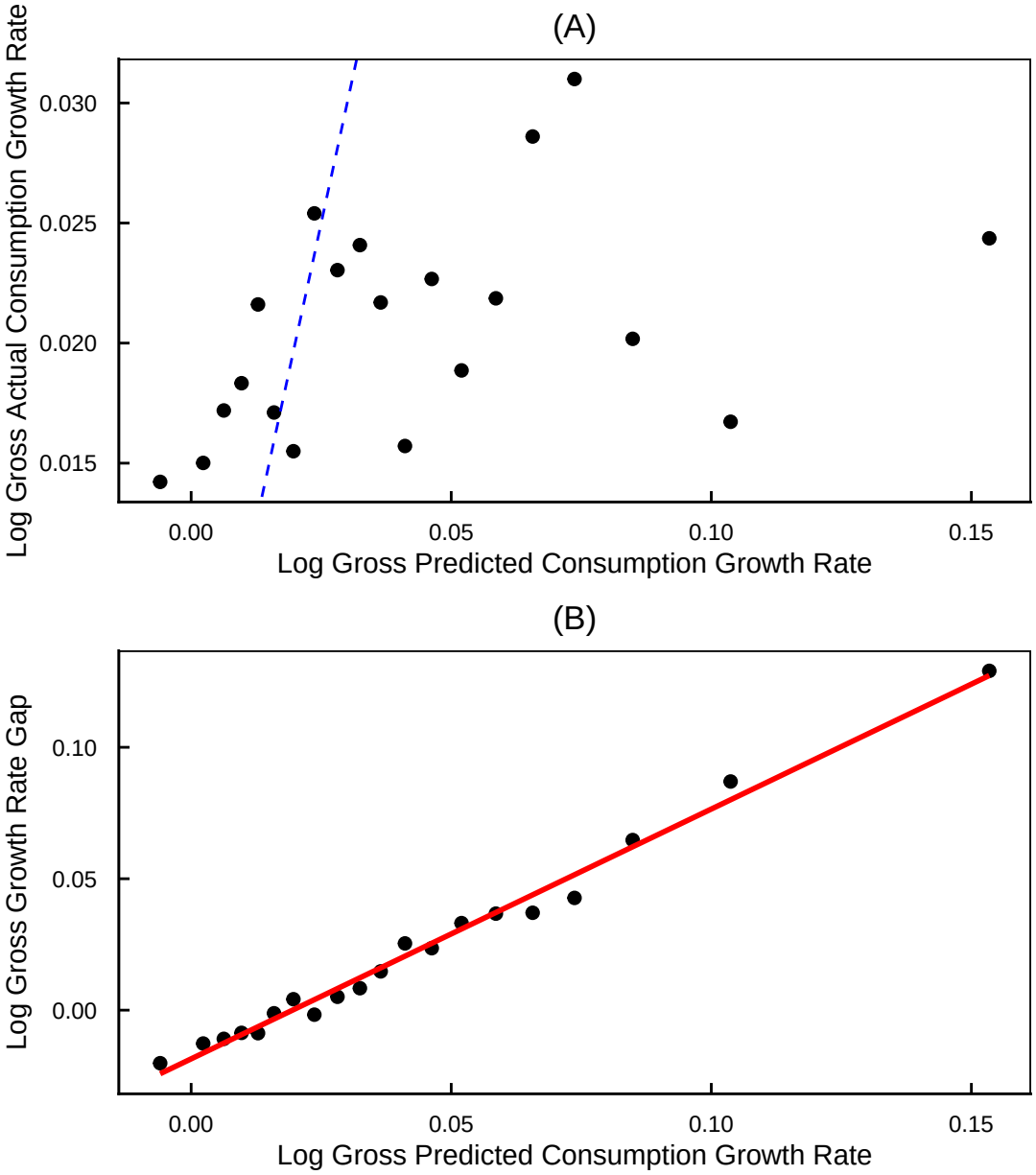
Notes: Red line is OLS fit. Panel data from 1950 to 2023. Capital stock in millions US$ at 2021 PPPs. Output−side real GDP in millions US$ at 2021 chained PPPs. Source: PWT 11.0.

countries. The returns are likely higher in regional levels and much higher in lower-income countries with lower public capital stocks. However, Besley and Persson (2013) show likely political economy reasons including political instability, favourtism, inequality, or corruptions which undermine the taxation-to-government-spending mechanism necessarily for optimal public capital investment. These factors are more prominent in lower-income countries, while countries might improve taxation capacities and have reduced spending frictions over development.

However, if public capital investments yield high returns, why would rational households even not pay taxes voluntarily but choose to avoid them when possible? Indeed, findings in Mauro (1995) also show low public (and private) investment levels when governments are inefficient, and Figure 3 shows that government effectiveness is lower in lower-income countries. Therefore, this high government spending friction could discourage households to pay taxes if they believe the revenue is not put in use efficiently. Nonetheless, Figure 4 shows that the trend in Figure 2 Panel (A) persist even controlling for government effectiveness. My following model will present that the household level, the fiscal features of lower-income countries can also create a collective action problem so that they optimally avoid paying taxes individually, thereby producing a global sub-optimal outcome.

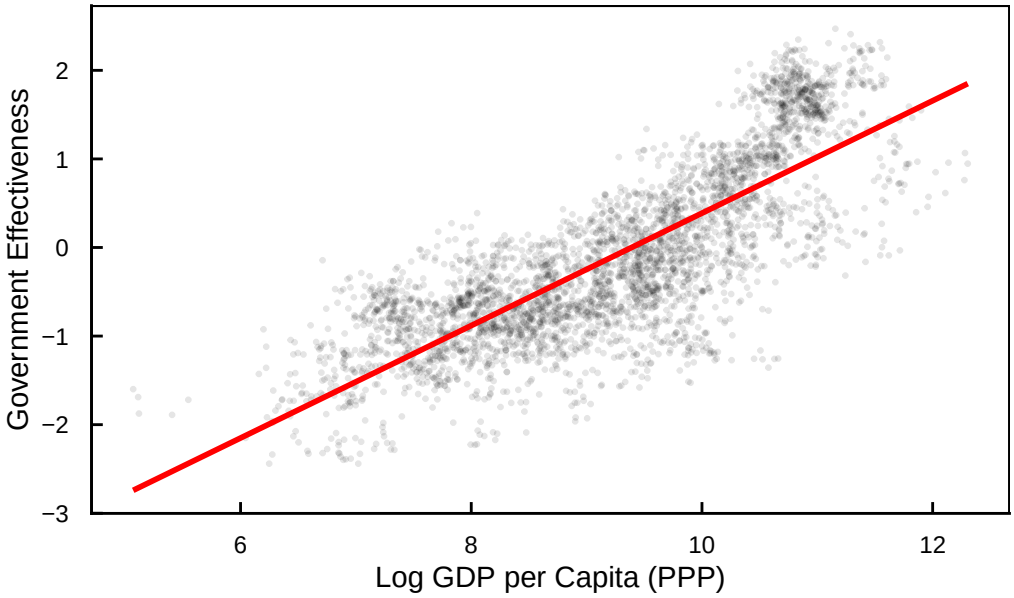
To illustrate, I will present an extreme case where we assume capitals are all publicly invested through taxation by governments. Consider the same optimisation problem as Equation 1 with a different budget constraint for homogeneous households in an economy. For any household i, the

Figure 2: Actual vs. Predicted Consumption Growth



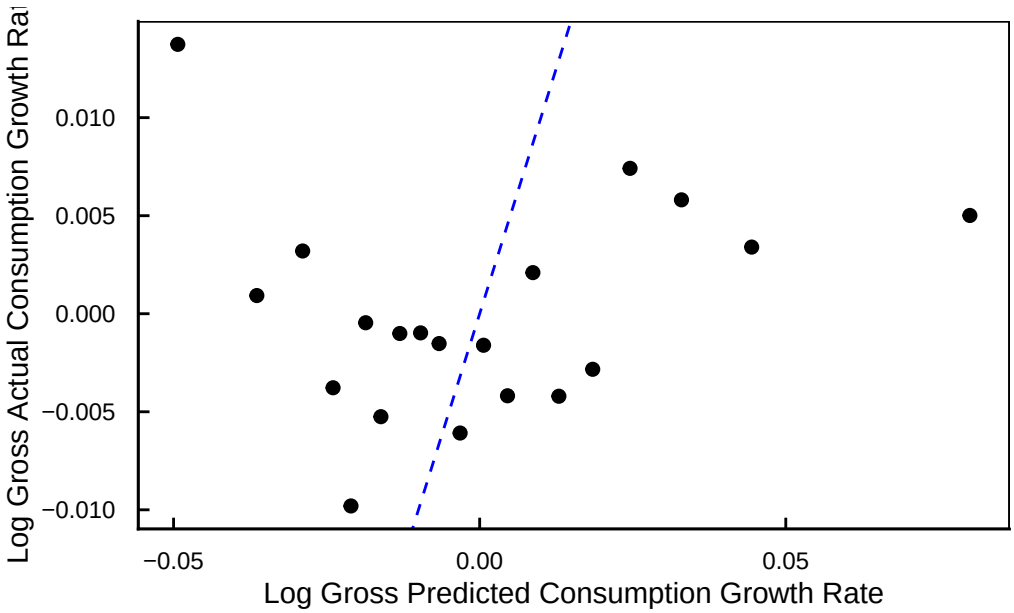
Notes: Binned scatter plots with 20 equal−frequency bins. The dashed line is the 45 degree line. The red line is OLS fit. Countries with populations below 1 million and irregular internal rates of return excluded. Data after 2019 is excluded to avoid Covid−19 distortions. Source: PWT 11.0.

Figure 3: Lower Government Effectiveness in Lower-Income Countries



Notes: Red line is OLS fit. Panel data from 1996 to 2023. Output−side real GDP in millions US$ at 2021 chained PPPs. Source: WDI.

Figure 4: Actual vs. Predicted Consumption Growth Controlled for Government Effectiveness



Notes: Binned scatter plots with 20 equal−frequency bins after partialling out government effectiveness. Source: PWT 11.0, WGI.

budget constraint is:

$$x_{i,t+1} = \frac{1}{N} \cdot f'(K_t) \cdot [(x_{i,t} - C_{i,t}) + \sum_{\forall j \neq i} (x_{j,t} - C_{j,t})], \tag{5}$$

where N is the total number of households in the economy, and $\sum_{\forall j \neq i}(x_{j,t} - C_{j,t})$ is the aggregate public investment from all other households. Because public capitals can only be invested collectively, the total returns depend on the aggregate investment level. For simplicity, assume each individual household receive an equal share of the total returns, public capital investment is the only government expenditure, and the government has perfect efficiency.

Weak Taxation Capacity

First consider a scenario where the state taxation capacity is weak. To reflect weak taxation capacity, assume that household i makes public investment decision through taxation voluntarily and independent of other households' decisions. From the perspective of household i, the total returns from public capital investment as a public good is diluted among the population. Alternatively, for a fixed level of total returns on public capital, the impact of the household's investment level $(x_{i,t} - C_{i,t})$ on its assets in the next period is small for large N.

Solving for optimal intertemporal consumption path for household i given $\sum_{\forall j \neq i}(x_{jt} - C_{jt})$:

$$u'(C_{i,t}) = \frac{1}{N} \cdot \delta f'(K_t) u'(C_{i,t+1}). \tag{6}$$

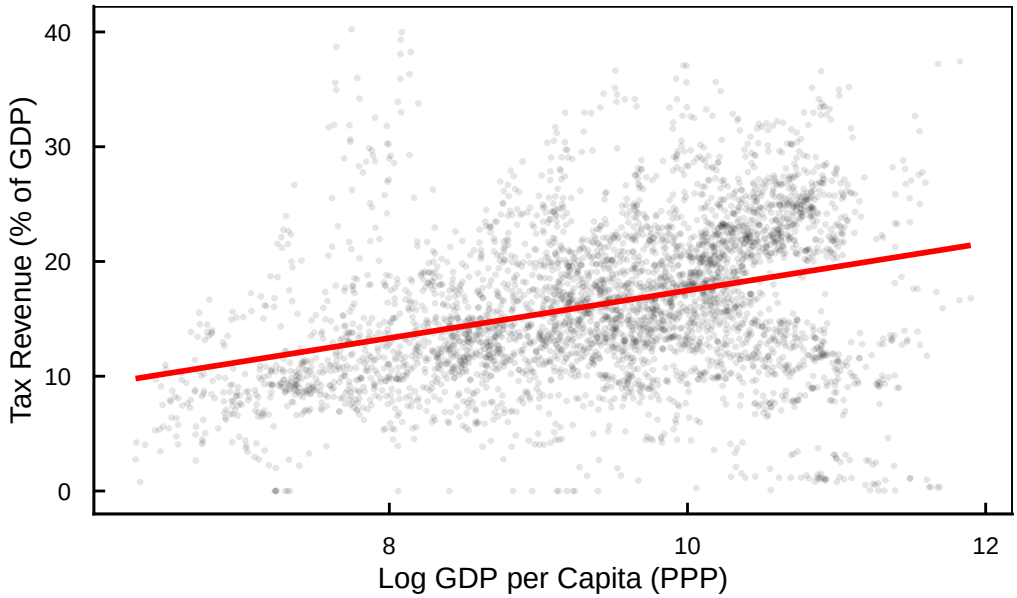
Notice that when $N = 1$, this equation is identical to Equation 3, but $u'(C_{i.t})$ decreases as N becomes larger. For increasing and concave utility function, $u'(C_{i,t})$ is smaller when $C_{i,t}$ is larger. This means that household i will consume more and invest less as N increases. For sufficiently large N so that the Euler Equation cannot be satisfied by an interior solution, household i will optimally choose a corner solution of consuming all and invest none. $\frac{1}{N}$ produces a collective action problem where the global optimal decision is for all households to invest and benefit from high $f'(K)$ but individuals households have a dominant strategy of not investing.

Figure 5 shows the countries with lower GDP per capita have lower tax revenue relative to their GDP. Therefore, this scenario is more consistent with lower-income settings and it is consistent with the reluctant tax compliance attitude among many low- and middle-income settings. This framework also provides an explanation to the prima facie irrationality of tax avoidance in lower-income countries where people can benefit largely from public investments.

Strong Taxation Capacity

Higher income countries tend to have stronger taxation capacity because, for example, income tax can be more easily collected from employees (Jensen, 2022). Although, public investment is less voluntary for tax rates as given, households might make public investment decisions by influencing

Figure 5: Higher Tax Revene in Higher-Income Countries



Notes: Red line is OLS fit. Panel data from 1972 to 2023. Output−side real GDP in millions US$ at 2021 chained PPPs. Source: WDI.

the tax rates through political channels. The key difference is that they are more likely to expect other households to also comply with the decided upon tax rates.

Consider a simple case of homogeneous households facing the same optimisation problem as Equation 1 but assume instead perfect enforcability upon any chosen taxation level and perfect government efficiency. Any individual household i chooses a taxation level (public investment level) $(x_{i,t} - C_{it})$ and expect symmetry from other households. Because households are homogeneous, the following should hold:

$$[(x_{i,t} - C_{i,t}) + \sum_{\forall j \neq i} (x_{j,t} - C_{j,t})] = N \cdot (x_{i,t} - C_{i,t}),$$

so that the budget constraint becomes identical to Equation 2:

$$x_{i,t+1} = \frac{1}{N} \cdot f'(K_t) \cdot N \cdot (x_{i,t} - C_{i,t}) = f'(K_t)(x_{i,t} - C_{i,t}).$$

Solving for optimal consumption path will produce an Euler Equation identical to Equation 3. Notice that the collective action problem is resolved in this setting because the perfect tax enforcement enables households to optimise consumption conditional on the expectation of symmetrical behaviours from others.

2.3 A Generalised Model

The collective action problem explored above is an channel embedded in the structure of public capital investment, but the extent varies and likely interact with other public capital investment side factors. This means the real functional form is ambigous and might involve complicated interaction and recurrence structures. However, the public capital investment frictions at the household-decision level can be generalised as a wedge $\phi \in [0, 1]$ where lower values indicate larger frictions. This produce a generalised intertemporal household model with both private and public capital returns with the following budget constraint:

$$x_{t+1} = r_t^{eff} \cdot (x_t - C_t), \tag{7}$$

where

$$r_t^{eff} = (1 - \tau) \cdot f'(K_t^{priv}) + \tau \cdot \phi \cdot f'(K_t^{pub}),$$

with K^{priv} denoting private capital stock and τ denoting the share of the total investment amount that the household chooses to invest in public capital. Notice that households choose τ freely to maximise their budget constraint. Because $f'(K_t^{priv})$ and $f'(K_t^{pub})$ are concave, this means that at equilibrium:

$$f'(K_t^{priv}) = \phi \cdot f'(K_t^{pub}) \tag{8}$$

Therefore, private capital investment would be systematically larger than public capital investment for $\phi > 1$.

Solving for optimal consumption path under the above constraint produces an Euler Equation:

$$u'(C_t) = \delta \cdot r^{eff} \cdot u'(C_{t+1}),$$

and the consumption growth rate is given by:

$$\frac{C_{t+1}}{C_t} - 1 = (\delta \cdot r^{eff})^{\frac{1}{\sigma}} - 1. \tag{9}$$

Recall the pattern illustrated in Figure 2 Panel (B): countries with higher predicted consumption growth rates tend to exhibit larger gaps between predicted and actual growth. Under the above model, this is expected. Countries with high predicted growth are lower-income countries with low capital stocks and high $f'(K)$.

3 Quantifying the Public Capital Investment Frictions

In this section, I will attempt three methods to quantify the public capital investment frictions across countries and time with aggregate data. The theoretical model features four structural parameters:

$$\Theta = \{\delta, \sigma, \tau, \phi\},$$

where δ is the annual discount factor, σ is the risk aversion coefficient and $\frac{1}{\sigma}$ is the elasticity of intertemporal substitution, τ is the share of total investment in public capital, and ϕ is the severity of the collective action problem for public capital investment. To discipline the model, I exogenously fix δ with data on depreciation rates from PWT, and $\sigma = 2$ proposed by Kremer et al. (2019) based on the literature. Although in the theoretical model, τ is chosen by the households, the public capital investment shares from ICSD can be interpretated as the equilibrium outcome of this choice.

I will also proxy for the returns on private and public capitals households face by derivation from this Cobb-Douglas production function facing a representative household in the economy:

$$Y_{it} = A_{it}(K_{it}^{priv})^{\alpha^{priv}}(K_{it}^{pub})^{\alpha^{pub}}L_{it}^{1-\alpha^{priv}-\alpha^{pub}},$$

where the first-order conditions are:

$$f'(K_{it}^{priv}) = \alpha_{priv} \cdot \frac{Y_{it}}{K_{it}^{priv}} \quad \text{and} \quad f'(K_{it}^{pub}) = \alpha_{pub} \cdot \frac{Y_{it}}{K_{it}^{pub}}.$$

Taking the conservative estimate of short-run national-level output elasticity of public capital 0.083 gives:

$$f'(K_{it}^{pub}) = 0.083\left(\frac{Y_{it}}{K_{it}^{pub}}\right)$$

Assume for simplicity the labour market is competitive in aggregate,

$$w = \frac{\partial Y}{\partial L} = (0.917 - \alpha^{priv})\frac{Y}{L},$$

so that:

$$\alpha^{priv} = 0.917 - \frac{wL}{Y},$$

where the PWT provides the data on the labour share of income. This approach would give a mean output elasticity of labour around 0.523, and a mean ouput elasticity of private capital around 0.394.

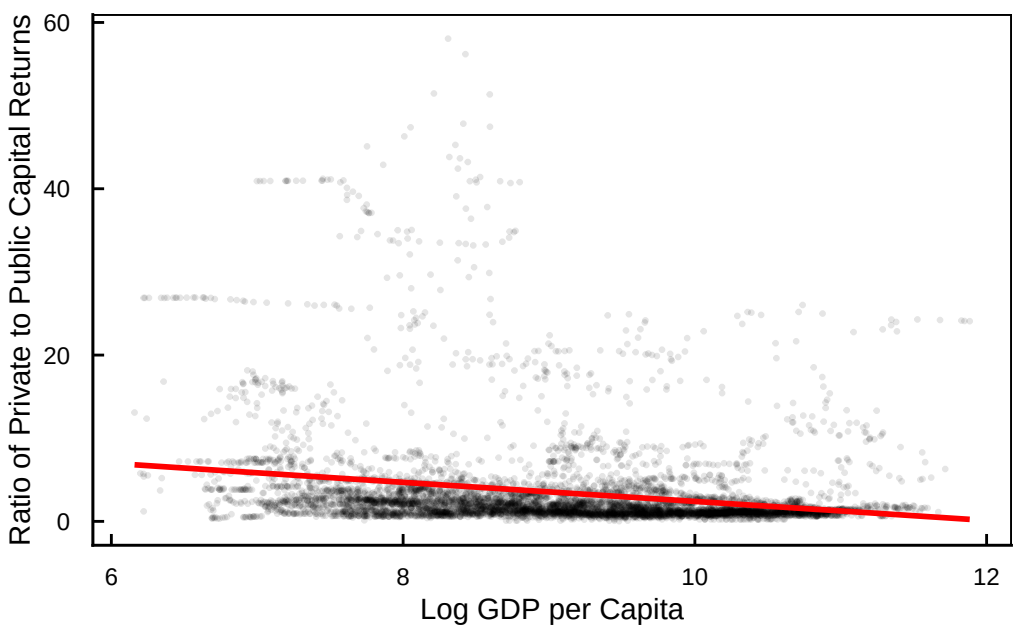
3.1 Difference in Private and Public Capital Returns

Recall the equilibrium condition of Equation 8, I can rewrite:

$$\phi_{it} = \frac{f'(K_{it}^{priv})}{f'(K_{it}^{pub})} = \frac{\alpha^{priv}}{\alpha_{it}^{pub}} \left(\frac{K_{it}^{pub}}{K_{it}^{priv}} \right)$$

Figure 6 shows with very noisy data that countries across time with lower per capita GDP is associated with larger ratio of private over public capital returns, and it is significantly larger than 1. According to the theoretical model, this would have signaled low public capital investment frictions in lower-income countries, opposite to the model intuition. However, notice that the theoretical model omits the private capital investment frictions which very likely exist, and likely larger than the public-side frictions from very large ratios in the figure. Therefore, this derivation exercise cannot effectively quantify the wedge in the theoretical model.

Figure 6: Ratio of Private to Public Capital Returns by Log GDP per Capita



Notes: Red line is OLS fit. Sources: PWT 11.0, ICSD.

3.2 Self-Employment Share as A Proxy for ϕ

Instead, ϕ can be approximated by a variable that is differentially associated with public but not private capital investment, and Section 2.2 and Besley and Persson (2013) show that it should likely relate to government taxation capacities. The self-unemployment share in total employment can be a good proxy to consider in this context. Jensen (2022) identifies that the income tax exemption threshold co-moves with co-moves with the employee/self-employed composition over development, such that the tax base consistently covers segments where third-party enforceable employee income predominates (85–95% employee share). This is primarily because that taxing

self-employment is administratively more difficult. Therefore, in lower-income-countries where the self-employment sector is larger in the employment composition, the effective taxation capacity is lower. This pattern is consistent with Figure 7 Panel (A) which shows higher self-employment percentage in total employment associates with lower public investment levels since government expenditure relies heavily on taxation, controlling for country and year effects. An apparent intuition is that lower-income-countries have high tax exemptions and small tax bases to fund public capital investment because self-employment increases downwards along income, and governments target the tax-exemption thresholds so that self-employment-income is often exempt to protect tax morale(Jensen, 2022). Nonetheless, the intuition in collective action is still consistent with this pattern for explaining why tax compliance is difficult among the self-employed despite high returns in public capital, of which the governments target tax exemptions in response to. Although the pattern in Figure 7 Panel (A) could be endogenous due to factors such as institutions or public sector employment, Panel (B) shows the absence of a comparable relationship with private capital investment.[5] This suggests that the relevant channel operates specifically through public capital, consistent with the specification of ϕ in the theoretical model.

3.3 Structural Estimation by Grid Search

Using self-employment share as a proxy for ϕ and to allow different values across countries and time, I parameterise ϕ_{it} for country i in year t with the following logistic function to be bounded between 0 and 1:

$$\phi_{it} = \frac{1}{1 + \exp(\beta_0 + \beta_1 \cdot SelfEmp_{it})}$$

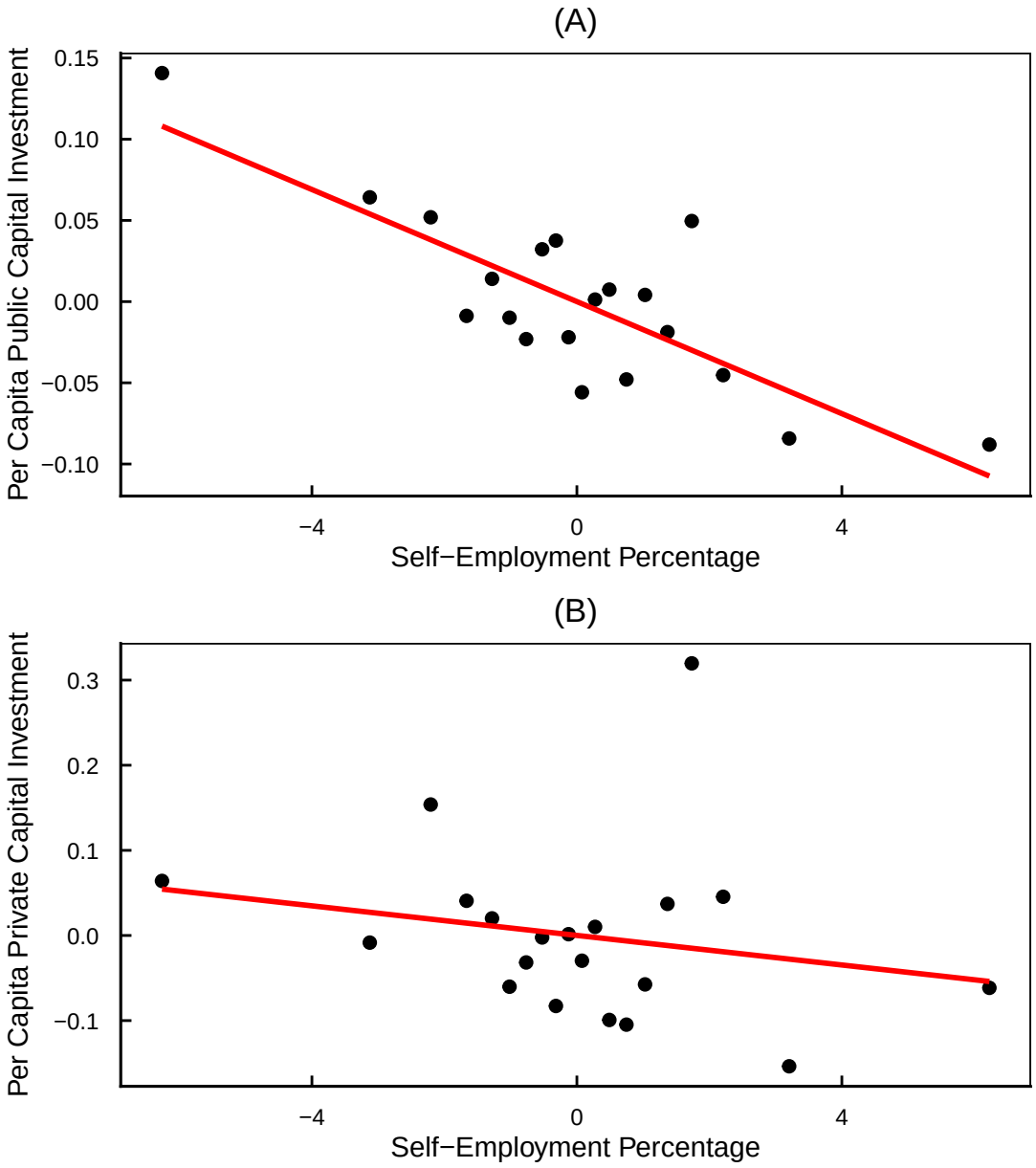
For estimation, I take logs on both sides of the Euler equation for a more tractable estimating equation:

$$\ln(\frac{C_{i,t+1}}{C_{i,t}}) = \frac{1}{\sigma} \ln(\delta \cdot r_{it}^{eff}(\beta)) + \varepsilon_{it},$$

where the residual term ε_{it} captures all other possible channels such as credit constraints or institutional features.

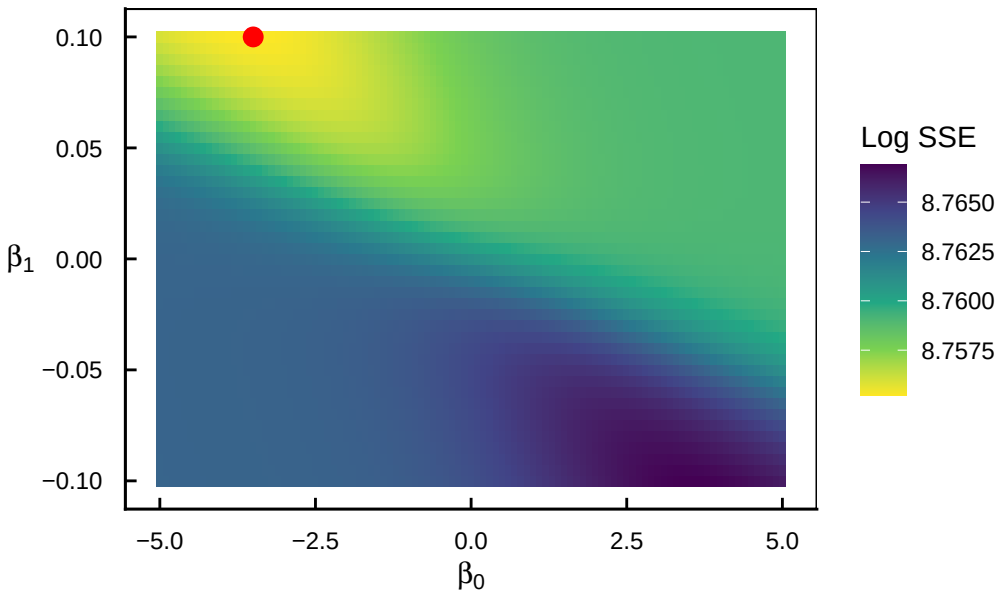
[5] $\beta = -0.021$, $p < 0.001$ for Panel (A); $\beta = -0.012$ $p = 0.064$ for Panel (B).

Figure 7: Public vs. Private Capital Investment and Self-Employment



Notes: Binned scatter plots with 20 equal−frequency bins after partialling out country and year fixed effects. Each point represents the within−bin mean. Red line is OLS fit. Investment in PPP international dollars (2017 ICP benchmark). Sources: ICSD, WDI.

Figure 8: Grid Search



Notes: Heatmap of log sum of squared residuals over grid of beta0 and beta1. Red dot indicates the parameter combination minimising SSE. Sources: PWT 11.0, ICSD, WDI.

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